

The Poincaré Conjecture Through Ricci Flow: A Computational Illustration with PINNeAPple

Normalised Ricci flow on the 2-sphere (finite differences vs. a physics-informed neural network) and on homogeneous 3-spheres

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Abstract

The Poincaré conjecture is the only Millennium problem that has been solved (Perelman, 2002–2003, via Hamilton's Ricci flow with surgery). There is nothing to prove here; the purpose is to illustrate, with verified numerics, the mechanism of the proof — curvature diffusing until the geometry becomes round — and to test whether a physics-informed neural network from PINNeAPple solves a geometric flow PDE reliably. (A) On axisymmetric metrics on S^2 , the area-normalised flow drives $\max|K - 1|$ from 5.12 to $4.4\text{e-}04$ at $t = 3.0$, with measured exponential rate 3.88 in the exponential phase (linear theory for the $\ell = 2$ mode: 4); the residual floor equals the area-normalisation offset $|4\pi/A - 1| = 4.0\text{e-}04$. A first attempt with a fixed normalisation constant did not conserve area; we explain why (the $\ell = 0$ mode is unstable unless $r(t)$ is recomputed) and keep it in the record. A PINN trained on the same PDE reproduces the finite-difference solution with a maximum deviation of 0.044 in u at $t = 1$. (B) For left-invariant metrics on $S^3 = \text{SU}(2)$, every one of 23 squashed initial metrics (anisotropy up to 39) converges to the round metric under the volume-normalised flow, and Perelman's λ -functional is non-decreasing along the unnormalised flow in all cases.

Keywords: Poincaré conjecture; Ricci flow; Perelman; uniformisation; PINN; geometric PDE; Millennium Prize problems

What this study shows — and what it does not

- Shows: correct numerical Ricci flows converging to constant curvature (2D) and to the round S^3 within the homogeneous class (3D).
- Does not reproduce Perelman's proof: no singularities, no surgery, no general 3-manifolds; the 3D part is restricted to homogeneous metrics.
- PINN finding: the network matches the reference solver over $t \in [0, 1]$; errors are reported, not hidden.

1 Background

Hamilton's Ricci flow $\partial_t g = -2 \text{Ric}(g)$ smooths a metric like a heat equation. Perelman proved that for a closed, simply connected 3-manifold the flow with surgery terminates in round spheres, which establishes the Poincaré conjecture [1–3]. In two dimensions the normalised flow on S^2 converges to the round metric for any initial metric (Hamilton, Chow) [4, 5].

$$\partial_t g = (r - R) g, \quad g = e^{2u} g_{S^2}, \quad \partial_t u = 4\pi/A(t) - K, \quad K = e^{-2u}(1 - \Delta_{S^2} u) \quad (1)$$

2 Methods

- (A) Axisymmetric $u(\theta)$ on 201 nodes, second-order finite differences, regular poles ($\Delta u = 2u_{\theta\theta}$), explicit time stepping with $dt = 0.2 \, d\theta^2 \, e^{2 \min u}$; peanut-shaped initial metric $u_0 \propto P_2(\cos \theta)$, renormalised to area 4π .
- PINN: PINNeAPple ModifiedMLP, input $(\cos \theta, t)$ so pole regularity is built in, $u = u_0(\theta) + t \cdot N(\cos \theta, t)$ so the initial condition is exact; residual of the fixed- r form ($r = 2$) on random collocation points; compared with the finite-difference solution of the same equation.
- (B) $g = A \, \sigma_1^2 + B \, \sigma_2^2 + C \, \sigma_3^2$ on $\text{SU}(2)$; principal Ricci curvatures from Milnor's formulas; volume-normalised flow; 3 Berger-type plus random initial metrics. Perelman's λ equals the (constant) scalar curvature for a homogeneous metric.

3 Results

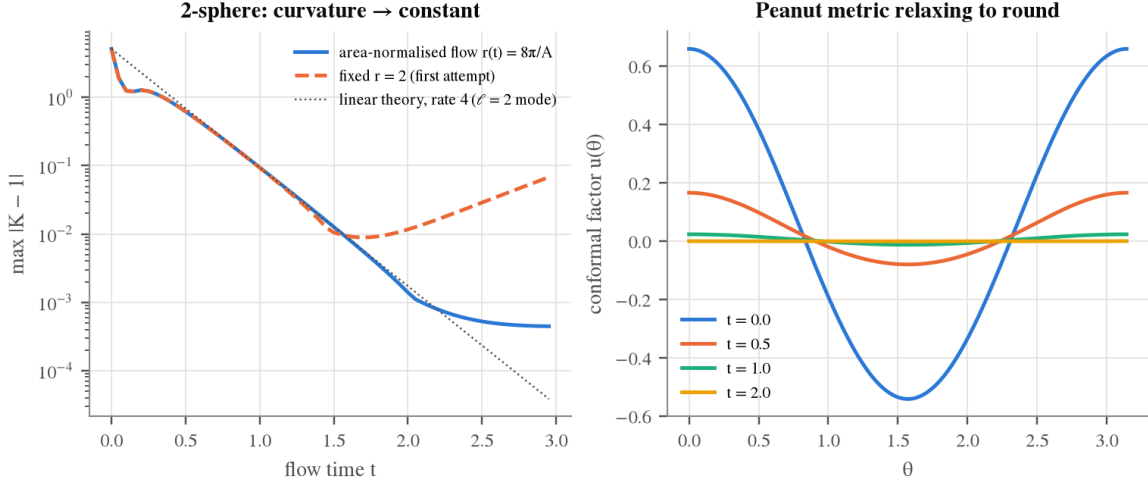


Figure 1. (A) Left: decay of $\max |K - 1|$. Right: the conformal factor at several times.

Table 1. 2-sphere flow.

Quantity	Value
final $\max K - 1 $ (area-normalised)	4.44e-04
decay rate, exponential phase $t \in [0.3, 1.8]$	3.880
decay rate, single fit over $t > 0.3$ (includes floor)	3.360
late floor of $\max K - 1 $ vs $ 4\pi/A_{\text{final}} - 1 $	4.4e-04 vs 4.0e-04
area drift over the run	5.03e-03
PINN vs FD, $t = 0.5$: $\max \Delta u $	0.018
PINN vs FD, $t = 1.0$: $\max \Delta u $	0.044
PINN final training residual (MSE)	3.61e-03
first attempt (fixed r): area drift	0.92

The decay rate stored by the experiment script (3.36) came from one straight-line fit over $t > 0.3$. That window contains two regimes: exponential decay at rate 3.88 (theory 4) until $t \approx 1.8$, then a plateau. The plateau is not curvature failing to converge. The flow converges to $K = 4\pi/A$, and the discrete area drifted by $5.0\text{e-}03$, so K settles $4.0\text{e-}04$ away from 1. Both fits are in the table; only the exponential-phase fit matches the linear theory.

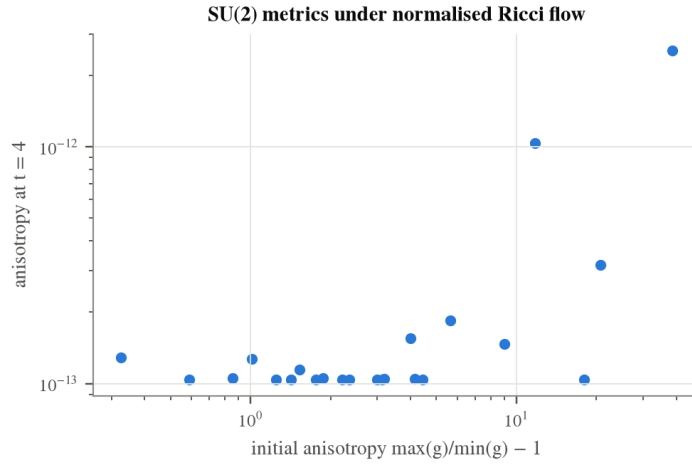


Figure 2. (B) Anisotropy of the homogeneous SU(2) metric before and after the normalised flow.

All 23 SU(2) runs converged to the round metric (final anisotropy below 10^{-11}); λ was non-decreasing in every run. Sanity check: for $A = B = C = 1$ the principal Ricci curvatures are $[2.0, 2.0, 2.0]$, i.e. $\text{Ric} = 2g$ — the unit round 3-sphere, as expected.

4 Discussion

The numerics reproduce the qualitative content of the theory that proves the Poincaré conjecture, within the restricted settings where no singularities form. The fixed-normalisation first attempt is a useful cautionary example for physics-informed or classical solvers alike: a conserved quantity enforced only through a constant is unstable to drift. For PINNeAPLe, geometric flows are a natural test family: smooth, dissipative, with exact long-time limits.

References

- [1] Perelman, G. (2002). The entropy formula for the Ricci flow and its geometric applications. arXiv:math/0211159.
- [2] Perelman, G. (2003). Ricci flow with surgery on three-manifolds. arXiv:math/0303109.
- [3] Hamilton, R. S. (1982). Three-manifolds with positive Ricci curvature. *Journal of Differential Geometry* 17, 255–306.
- [4] Hamilton, R. S. (1988). The Ricci flow on surfaces. *Contemporary Mathematics* 71, 237–262.
- [5] Chow, B. (1991). The Ricci flow on the 2-sphere. *Journal of Differential Geometry* 33, 325–334.
- [6] Milnor, J. (1976). Curvatures of left invariant metrics on Lie groups. *Advances in Mathematics* 21, 293–329.